

Insider Threat & Data Exfiltration Investigation

Analysis of suspected KYC data exfiltration and anti-forensic activity on a Windows 10 Enterprise LTSC system

CASE REFERENCE

ITI-2026-001

EXAMINER

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REPORT DATE

July 12, 2026

REPORT VERSION

1.0 — Final

CLASSIFICATION

Academic Portfolio

TARGET SYSTEM

DESKTOP-2F965T3

EVIDENCE IMAGE (MD5)

7e5404e298b1d886534a2845e4d880bc

STANDARDS REFERENCED

ISO/IEC 27037, 27035, 27041, 27042, 27043 · NIST SP 800-86, 800-61 · RFC 3227 · LGPD

This report documents a digital forensic investigation conducted in a controlled laboratory environment for academic and portfolio purposes. The target organization "NexChain Exchange", the user accounts, and all customer data referenced herein are fictitious. All findings, methodologies, and conclusions are the work of the examiner and reflect standard forensic practice. This document does not constitute a legal instrument in any jurisdiction.

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1. Identification

Case Reference	ITI-2026-001
Examiner	Paulo Vaz
Requesting Party	Academic Portfolio — Self-initiated
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2. Executive Summary

This report documents a digital forensic investigation conducted on a Windows 10 Enterprise LTSC virtual machine (host name DESKTOP-2F965T3, designated **NEXCHAIN-WS01** within the fictitious NexChain Exchange environment). The investigation was carried out in a controlled laboratory as part of an academic portfolio project.

The examination sought to determine whether a compliance analyst had exfiltrated customer KYC (Know Your Customer) data — names, national identifiers, and cryptocurrency wallet addresses — to an external device prior to resignation, and whether anti-forensic techniques were used to conceal the activity. The investigation was conducted as a **blind analysis**: the attacker's actions were recorded and sealed in a GPG-encrypted log before the investigation began, and that log was opened only after all findings were finalized, to allow an objective measure of investigative completeness (Section 11).

The investigation spanned eight phases, from attack simulation and forensic acquisition through memory, disk, and anti-forensic analysis to timeline reconstruction and this report. Multiple independent tools were employed — Volatility 3, The Sleuth Kit 4.14.0, and native forensic utilities — with cross-validation between evidence sources used as a methodological control.

Principal findings

Finding	Detail
Data exfiltration confirmed	A KYC dataset (<code>kyc_customers.csv</code>) was copied from the user's Documents folder to a mounted virtual disk (drive E:) acting as a simulated removable device.
Three anti-forensic techniques used	Secure deletion (<code>SDelete</code>), Security event-log clearing (<code>wevtutil</code>), and NTFS timestamp manipulation (timestamping) were all executed and all independently detected.
Full command history recovered	The complete PowerShell command history of both user accounts (<code>analyst01</code> and <code>admin</code>) was recovered from disk, providing a verbatim, first-party record of every action.
Simulated device identified as virtual	Registry analysis established the "USB device" was a Microsoft Virtual Disk attached internally via the OS, not a physical USB device — a distinction proven through the SCSI enumeration vendor string.
Log clearing self-recorded	The act of clearing the Security log generated Event ID 1102, which itself records the responsible account (<code>admin</code>) and precise time — the attacker could not suppress this record.
Timestamping structurally proven	A discrepancy between the NTFS <code>\$STANDARD_INFORMATION</code> and <code>\$FILE_NAME</code> timestamps conclusively demonstrated timestamp falsification, independent of any log.
100% investigative coverage	Every action recorded in the sealed attacker log was independently recovered by the investigation, most through multiple convergent sources (Section 11).

3. Objectives

The investigation was conducted with the following objectives:

1. Determine whether customer KYC data was copied from the target system to an external device.
2. Identify and recover all disk and memory artifacts related to the suspected exfiltration and any anti-forensic activity.
3. Determine the nature of the external device used (physical vs. virtual).
4. Detect and technically prove any use of anti-forensic techniques (secure deletion, log tampering, timestamp manipulation).
5. Reconstruct a unified, chronologically consistent timeline correlating memory, disk, and event-log evidence.
6. Assess investigative completeness through blind comparison against a pre-sealed record of the attacker's actions.
7. Produce a formal forensic report documenting all findings, methodology, limitations, and procedural deviations.

4. Methodology & Standards

The investigation followed a structured eight-phase methodology aligned with international standards for digital evidence handling and incident investigation. Each standard governs a specific dimension of the work:

Standard	Application in this investigation
ISO/IEC 27037	Identification, collection, acquisition and preservation of digital evidence — governs the forensic imaging and hashing procedures in Phase 02.
ISO/IEC 27035	Information security incident management — frames the insider-threat scenario as a security incident with a structured response.
ISO/IEC 27041	Assurance for the suitability and adequacy of the investigative method — supports the multi-tool cross-validation approach.
ISO/IEC 27042	Analysis and interpretation of digital evidence — governs the memory, disk, anti-forensic, and timeline-correlation analysis.
ISO/IEC 27043	Incident investigation principles and processes — governs the end-to-end investigative structure.
NIST SP 800-86	Guide to integrating forensic techniques into incident response — informs the acquisition-to-analysis workflow.
NIST SP 800-61	Computer security incident handling guide — informs the incident-response framing.
RFC 3227	Guidelines for evidence collection and archiving — the <i>order of volatility</i> it defines justifies capturing volatile memory before shutdown, prior to disk acquisition.
LGPD	Brazil's General Data Protection Law — establishes the regulatory significance of the exfiltrated KYC data as personal data (see Sections 5 and 11).

Phase structure

Phase	Summary
Phase 00 — Environment Setup	Two-VM laboratory: Windows 10 Enterprise LTSC target and Kali Linux 2026.1 investigator, connected via host-only network; clean baseline snapshot taken before any activity.
Phase 01 — Attack Simulation	Controlled generation of the incident from the attacker's perspective; actions sealed in a GPG-encrypted log for later blind comparison.
Phase 02 — Forensic Acquisition	Memory dump captured pre-shutdown (order of volatility, RFC 3227); disk acquired to E01 via <code>ewf_acquire</code> with MD5 + SHA-256 verification.
Phase 03 — Memory Analysis	Volatility 3 analysis of the 8.7 GB memory image; recovery of process list and command-history fragments.
Phase 04 — Disk Analysis	The Sleuth Kit analysis of the E01: MFT, USN Journal, Prefetch, Event Log, Registry, and NTFS timestamp analysis.
Phase 05 — Anti-Forensic Analysis	Evaluation of each anti-forensic technique's effectiveness against what the investigation recovered.

Phase 06 — Timeline Reconstruction	Correlation of all timestamped events across sources into one UTC-normalized timeline.
Phase 07 — Report & Sealed Log Comparison	Blind-analysis comparison and this report.

5. Case Description

The target system is a Windows 10 Enterprise LTSC 21H2 virtual machine (build 10.0.19044) with 8 GB RAM, 4 virtual CPUs, and a 50 GiB NTFS system partition. Two local user accounts are relevant to the investigation:

analyst01	Standard user — represents the suspected insider (a compliance analyst). SID S-1-5-21-399538211-3085301091-319685644-1003.
admin	Local administrator — used for actions requiring elevation (virtual disk creation, event-log clearing).

The scenario models a compliance analyst at the fictitious NexChain Exchange suspected of exfiltrating customer KYC data prior to resignation. The KYC dataset contained names, Brazilian national identifiers (CPF), and cryptocurrency wallet addresses — a category of personal and financial data whose unauthorized disclosure falls within the scope of the **LGPD**. The analyst allegedly copied the data to an external device and attempted to conceal the activity using anti-forensic techniques.

The investigation was initiated as an academic exercise demonstrating end-to-end DFIR methodology — from evidence generation and forensic acquisition through memory and disk analysis to formal reporting — using real forensic tools and industry-standard procedures.

6. Evidence & Chain of Custody

6.1 Primary Disk Image

Description	Windows 10 Enterprise LTSC 21H2 system disk (post-attack state)
Source	Consolidated clone of the "post-attack" VirtualBox snapshot
Image format	Expert Witness Format (E01), 35 segments
Logical size	50 GiB (53,687,091,200 bytes)
Acquisition tool	ewf_acquire 20140816
Acquisition date	July 6, 2026
MD5 (stored / verified)	7e5404e298b1d886534a2845e4d880bc (match)
SHA-256	3d61d7774ca4f85a88077d46ce535debe463f341aff2404cc885dc3471886188
Partition offset (TSK)	104448 (primary NTFS)
Status	Active — authoritative image for all disk analysis

6.2 Memory Image

Description	Full RAM capture taken immediately before shutdown (order of volatility, RFC 3227)
File	memdump.elf (ELF core dump)
Acquisition tool	VBoxManage debugvm ... dumpvmmcore
Size	8,730,336,996 bytes (~8.7 GiB)
SHA-256	9e2a212c3a63e484cde27151467fcaa75b9a13aaf3945707b8c8bca2ce14a90e
Status	Secondary evidence — corroborates disk findings

6.3 Sealed Attacker Log

The attacker's actions were documented in real time during Phase 01 and immediately sealed with GPG (AES-256) symmetric encryption before the investigation began. The encrypted file remained sealed until all investigative findings were finalized, at which point it was decrypted for the blind-analysis comparison in Section 11. This procedure preserves investigator objectivity and provides an auditable measure of investigative completeness.

Procedural note — acquisition destination. Two initial disk-acquisition attempts failed mid-write due to I/O instability in the VirtualBox shared-folder (vboxsf) layer used as the initial output target. The acquisition was successfully completed to a dedicated, locally-attached virtual disk instead. No evidence was lost or altered; the failed partial files were discarded before the successful acquisition. Full detail is recorded in the chain-of-custody

document. This deviation would warrant a certified hardware write-blocker and dedicated evidence storage in a production context.

7. Forensic Timeline

The following unified timeline was reconstructed from independent evidence sources spanning memory, disk metadata, and the Windows event log. **All timestamps are normalized to UTC**, the standard reference frame for forensic reporting, so that events originally recorded in different time zones (host log in BRT/UTC-3; system artifacts in UTC; TSK output in EDT/UTC-4) sit on one comparable axis.

UTC	Original	Zone	Event	Source	Type
~14:45	11:45	BRT	VHD created and mounted as E:	Sealed log	attacker
14:50:45	10:50:45	EDT	kyc_customers.csv created in Documents	\$FN via istat	system
~15:06	12:06	BRT	File copied to E:\ (exfiltration)	Sealed log	attacker
~15:16	12:16	BRT	SDelete run on original file	Sealed log	attacker
15:20:50	15:20:50	UTC	PowerShell session #1 started	pslist (memory)	system
15:20:54	15:20:54	UTC	Security log cleared — Event ID 1102	Event Log	system
15:26:06	15:26:06	UTC	PowerShell session #2 started	pslist (memory)	system
15:26:30	11:26:30	EDT	Timestomping applied (MFT Modified)	istat \$SI	attacker
16:10:21	16:10:21	UTC	Memory dump captured (pre-shutdown)	windows.info	investigator

Note on time-zone normalization. Artifacts from the same Windows system surface here in different zones (Event Log in UTC, TSK istat in EDT) — this reflects how different subsystems store and how different tools render time, not a clock inconsistency. The zone conversions were validated using a cross-source anchor: the `wevtutil` action appears in the hand-written log at 12:21 (BRT) and in Event ID 1102 at 15:20:54 (UTC), a ~3-hour offset confirming BRT = UTC-3. Times marked "~" derive from the manually-written attacker log and are approximate to the minute; where they diverge from system-generated timestamps, the system source is authoritative.

Reading the timeline as a narrative, the operation reconstructs cleanly: preparation (VHD creation) → data staging (KYC file creation) → exfiltration (copy to E:) → a concentrated anti-forensic phase (SDelete, log clearing, and timestomping within a ~10-minute window, 15:16–15:26 UTC) → and finally the investigator's memory capture. The concentration of all three anti-forensic techniques into a single short window is itself an analytical indicator of a deliberate, staged clean-up rather than scattered, reactive concealment.

8. Technical Analysis

8.1 Memory Analysis (Phase 03)

The 8.7 GB memory image was analyzed with Volatility 3 Framework 2.28.1. The system profile was confirmed as Windows 10 build 19041, consistent with the disk image.

Process list (windows.pslist). Two distinct PowerShell sessions were active at capture time: one started at 15:20:50 UTC (spawned by a RuntimeBroker process) and one at 15:26:06 UTC (spawned by explorer.exe — the expected parent for a manually opened terminal). Neither sdelete64.exe nor wevtutil.exe appeared in the process list, consistent with both having already exited by capture time — expected for short-lived command-line tools.

Command history in memory. A raw string search of the memory image recovered fragments of the attacker's interactive PowerShell history — including the literal SDelete command line and the file paths involved — from the PSReadLine console buffer retained in RAM. This established, from volatile memory alone, that the deletion and copy operations had occurred, even though the responsible processes were no longer running and the original file had been removed from disk.

File scan. A single file-object reference to kyc_customers.csv remained resident in memory, consistent with the copy present on the simulated USB device (E:) at capture time.

Network state (windows.netstat). Three established TCP connections were present, all on the NAT interface and directed to Microsoft Azure address ranges (routine telemetry/update traffic). No evidence of network-based exfiltration was found — consistent with the physical-media exfiltration method established elsewhere.

8.2 Disk Analysis (Phase 04)

The E01 image was mounted read-only at every layer and analyzed with The Sleuth Kit 4.14.0.

8.2.1 Deleted-file recovery and the USN Journal

The original kyc_customers.csv in the user's Documents folder was not recoverable via the Master File Table (MFT): three progressively broader `fts` searches returned no entry — not even a deleted or orphaned record. This is a stronger anti-forensic result than a standard deletion, which typically leaves a recoverable MFT entry; SDelete's secure-overwrite behavior invalidated the entry itself.

However, the file's prior existence was proven through the **USN Journal** (`$Extend\$UsnJrnl:$J`), an NTFS change-log mechanism architecturally separate from the MFT. A UTF-16LE string search of the extracted journal returned **ten occurrences** of kyc_customers.csv, confirming the file existed and underwent file-system activity despite being unrecoverable from the MFT.

8.2.2 Prefetch — tool execution

Windows Prefetch caches an entry for every executed program, independent of whether the process still exists. Two entries confirmed execution of both anti-forensic tools:

Prefetch file	Last execution (local)
SDELETE64.EXE-6548E1FA.pf	2026-07-04 11:11
WEVTUTIL.EXE-EF5861C4.pf	2026-07-04 11:20

8.2.3 Windows Event Log — the self-recording log clear

The `Security.evtx` log was extracted and parsed. Its small size and the fact that its first surviving event is the clearing action itself both indicate the log was reset. That clearing action is recorded by **Event ID 1102** — an event Windows generates automatically whenever the audit log is cleared, which cannot be suppressed by the clearing command itself:

Event ID	1102 — "The audit log was cleared"
Timestamp	2026-07-04 15:20:54.560593 UTC
Responsible account	admin
Client Process ID	7812

Process-creation auditing (Event ID 4688) was not enabled in the target's default configuration, so SDelete's process creation was not captured in the Security log. This is an expected consequence of the default Windows audit policy and is noted as an environmental limitation rather than an investigative gap; execution was independently confirmed via Prefetch and command history.

8.2.4 Registry — nature of the external device

The nature of the "USB device" was established through a three-key elimination process in the SYSTEM hive:

- **USBSTOR** — this key did not exist. It is populated only when a device enumerates over the USB bus, so its absence rules out a physical or passed-through USB device.
- **MountedDevices** — recorded drive letter `E:` mapped to a disk signature distinct from the CD-ROM/optical entries, confirming `E:` was a real, separately-attached disk.
- **SCSI enumeration** — contained the entry `Disk&Ven_Msft&Prod_Virtual_Disk`. The vendor string "Msft" (Microsoft) identifies this as a virtual disk created by the Windows Virtual Disk Service (the mechanism behind `diskpart create vdisk`), in contrast to the VM's real disk which enumerates as `Ven_VBOX`.

The combination — absent from USB, present in SCSI under a Microsoft vendor string — conclusively identifies the exfiltration medium as a virtual disk (VHD) attached internally via the operating system, not a physical USB device.

8.2.5 PowerShell command history — the primary artifact

PowerShell's `PSReadLine` module persists console input history to disk, per user profile. Both relevant profiles retained complete histories:

analyst01 — the full attack session: SDelete download and setup, KYC file creation (including a first failed attempt preserved verbatim), the copy to `E:`, the SDelete execution, and the complete three-line timestomping sequence.

admin — two lines: `diskpart` (virtual disk creation) and `wevtutil cl Security` (log clearing).

This is the single strongest artifact in the investigation. Unlike volatile memory (lost on shutdown) or the USN Journal (which proves activity but not the exact command), this is a persistent, unmediated, first-party record — the shell recording exactly what was typed. Because each account maintains its own history, actions could be attributed to a specific account — establishing not merely *what* occurred but *who* performed each step, which is central to an insider-threat case.

8.2.6 NTFS timestamp analysis — timestomping

NTFS maintains two independent timestamp sets per file: **\$STANDARD_INFORMATION** (\$SI), modifiable via standard Windows APIs and displayed by Explorer and PowerShell; and **\$FILE_NAME** (\$FN), stored in the parent

directory index and not exposed to the same modification path. Timestamp-manipulation tools typically alter \$SI while leaving \$FN untouched, producing a detectable discrepancy. Examination of the exfiltrated file (via `istat`) showed exactly this:

Attribute set	Created	Interpretation
\$STANDARD_INFORMATION	2024-01-01 06:00:00	Falsified value planted by the attacker
\$FILE_NAME	2026-07-04 10:50:45	Real creation time, untouched

The discrepancy is conclusive, structural proof of timestamp falsification — and, critically, it does not depend on any log, memory image, or command history surviving. Additionally, the \$SI "MFT Modified" sub-field (2026-07-04 11:26:30) retained the real time of the timestomping action, because that field is updated automatically by the file system and was not settable via the technique used — providing a further corroboration point.

9. Anti-Forensic Analysis

Three anti-forensic techniques were used. In every case the technique achieved its narrowest data-level objective but failed to conceal that the activity had occurred, because each was independently detected through multiple convergent sources.

Technique	Attacker's goal	What succeeded	What exposed it
SDelete	Prevent recovery of the original file	MFT-level recovery blocked (content + entry)	USN Journal, Prefetch, PowerShell history, memory strings
wevtutil cl Security	Erase the audit trail	Prior log content destroyed	Event ID 1102 (self-recording), Prefetch, PowerShell history
Timestomping	Disguise the file's true date	\$SI (user-facing) successfully falsified	\$FN retained real time; \$SI MFT-Modified sub-field; PowerShell history

Overall assessment. All three techniques achieved their literal objective (destroy this specific data point) but none achieved the broader goal of concealing that anti-forensic activity took place. In each case at least three independent artifact categories — spanning memory, disk metadata, and persistent command history — converged on the same conclusion. This pattern is consistent with tooling designed to alter a single specific data point rather than comprehensively erase every trace of its own use across the operating system. Notably, the log-clearing technique is arguably counter-productive for the attacker: it destroys prior content but generates Event ID 1102, which draws direct attention to the exact moment concealment began.

10. Evidence Reliability Assessment

Not all evidence carries the same evidentiary weight. Following standard forensic classification, each item is graded as **primary** (a direct, first-party record of the act itself, requiring minimal interpretation), **corroborative** (independently supports a conclusion but is not, alone, the act being proven), or **circumstantial** (consistent with a conclusion but does not, by itself, establish it). This classification — recommended practice under **ISO/IEC 27041** (assurance of method and result reliability) — makes explicit which findings could stand alone under challenge and which depend on convergence with other sources.

Evidence	Classification	Basis for classification
PowerShell command history (ConsoleHost_history.txt)	Primary	A verbatim, unmediated record written by the shell itself at the moment each command was typed — no reconstruction or interpretation by the examiner.
Event ID 1102 (Security log clear)	Primary	Automatically generated by the Windows audit subsystem, not by the examiner or by any tool under attacker control — a first-party system record of the act.
NTFS \$SI vs \$FN discrepancy (istat)	Primary	A direct structural read of file-system metadata; the discrepancy itself is the fact being established, not an inference from a related artifact.
USN Journal entries	Corroborative	Confirms the file existed and underwent activity, but does not by itself identify which command caused each entry — supports, rather than independently proves, the exfiltration/deletion narrative.
Prefetch (.pf files)	Corroborative	Confirms an executable ran and when it last ran, but not what arguments were used or who initiated it — strengthens but does not alone establish attacker intent.
Memory strings (Volatility)	Corroborative	Recovered via broad string search rather than structured parsing of a known data structure; consistent with the command history but reconstructed, not native.
Registry SCSI vendor string (Ven_Msft)	Corroborative	Strongly indicative of a virtual disk, but relies on interpreting a vendor-identification convention rather than a direct assertion of device type.
PowerShell session parent process (RuntimeBroker anomaly)	Circumstantial	Atypical but not independently conclusive of anything; noted for awareness, not relied upon in any finding.

The investigation's central conclusions — exfiltration, the three anti-forensic techniques, and their attribution to specific accounts — are each supported by at least one **primary** source, with multiple corroborative sources in every case. No conclusion in this report rests on circumstantial evidence alone.

11. Blind-Analysis Comparison (Sealed Log)

After all findings were finalized, the GPG-sealed attacker log from Phase 01 was decrypted and compared against what the investigation had independently recovered. The purpose of this comparison is to provide an objective, auditable measure of investigative completeness — what was found, what was missed, and why.

#	Attacker action (sealed log)	Independently recovered?	Corroborating sources
1	VHD mounted as E:	Yes	Registry (MountedDevices, SCSI), PowerShell history (admin)
2	File copied to E:\	Yes	USN Journal, PowerShell history (analyst01), memory
3	SDelete on original file	Yes	Prefetch, PowerShell history, USN Journal, memory
4	wevtutil cl Security	Yes	Event ID 1102, Prefetch, PowerShell history (admin)
5	Timestamping applied	Yes	\$SI vs \$FN (istat), PowerShell history (analyst01)

Result: 100% of the recorded attacker actions were independently recovered, most through multiple convergent sources. No action escaped detection entirely.

A measured view of the result

Complete recovery in this case should be read with appropriate context, not as evidence that anti-forensics is futile in general:

- **Evidence quality varied by action, even where detection was complete.** Some actions were confirmed by four independent sources; others by two or three. Against a more thorough adversary — one who also cleared the PowerShell history (via `clear-history` and deletion of `ConsoleHost_history.txt`) — several actions would have rested on weaker evidence, and the single strongest artifact in this investigation would have been absent.
- **The investigation had visibility only from the first recorded action onward.** The sealed log begins at VHD creation; the analysis cannot speak to reconnaissance, decision-making, or preparation that may have preceded it.
- **Technical evidence establishes acts, not motive.** The investigation proves *what* occurred, *when*, and by *which account*, with high confidence. It does not establish intent, whether the actor operated alone or under direction, or what was ultimately done with the exfiltrated data — a structural limit of any technical examination.

12. Legal Admissibility Context

While this report is an academic exercise and does not constitute a legal instrument (Section 16), the methodology applied throughout was deliberately chosen to reflect practices that support evidentiary admissibility in a real proceeding. Four factors are particularly relevant:

- **Integrity and non-repudiation of the source.** The disk image was acquired via a forensically sound method (ewf/acquire, EnCase format) with dual-hash verification (MD5 and SHA-256, both independently re-verified), and all subsequent analysis was performed against read-only mounts. This chain — acquire, hash, verify, analyze without altering the source — is the foundation courts and opposing experts typically scrutinize first; any doubt about source integrity can undermine an otherwise sound analysis.
- **Reproducibility.** Every finding in this report is tied to a specific, documented command against a specific, hashed evidence file. In principle, a second examiner with access to the same E01 and memory image could repeat every step in Sections 8–11 and arrive at the same result — a core requirement for evidence to withstand challenge (aligned with **ISO/IEC 27041**, assurance of method adequacy).
- **Multi-source corroboration.** As formalized in Section 10, the central conclusions of this case rest on primary evidence corroborated by multiple independent sources, rather than on a single artifact. This reduces reliance on any one tool's correctness and mirrors how expert evidence is typically strengthened for cross-examination.
- **Documented chain of custody and disclosed deviations.** Every procedural deviation from ideal practice (Section 14) — including the shared-folder acquisition failure — is disclosed rather than omitted. Undisclosed deviations, if later discovered, are far more damaging to admissibility than disclosed ones; transparency about limitations is itself a credibility-preserving practice.

Under Brazilian procedure, this kind of technical examination would typically inform a *laudo pericial* prepared by a court-appointed or party-retained expert (*perito*), governed by the Código de Processo Penal / Código de Processo Civil rather than by this report's academic format. The forensic discipline demonstrated here — hash-verified acquisition, reproducible steps, and multi-source corroboration — is the same discipline that would underpin such a formal instrument.

13. Conclusions

Based on the totality of evidence across memory and disk analysis, the following conclusions are established with high confidence:

- 1 Customer KYC data was exfiltrated to an external device.**
The dataset was copied from `C:\Users\analyst01\Documents` to drive E:, confirmed by the PowerShell command history (both in memory and on disk), the USN Journal, and a memory-resident file object.
- 2 The external device was a virtual disk attached via the operating system.**
Registry analysis (absence in `USBSTOR`; presence in `SCSI` as `Ven_Msft_Virtual_Disk`) establishes the medium was a VHD attached internally, not a physical USB device.
- 3 Three anti-forensic techniques were used and all were detected.**
Secure deletion, Security-log clearing, and timestomping were each independently confirmed; none concealed the fact that the activity occurred.
- 4 Individual actions are attributable to specific accounts.**
Per-user PowerShell histories attribute exfiltration, secure deletion, and timestomping to `analyst01`, and virtual-disk creation and log clearing to `admin`.
- 5 The timeline is internally consistent across all sources.**
Memory, disk metadata, and event-log timestamps correlate coherently once normalized to UTC; the zone conversion was validated against a cross-source anchor.
- 6 The incident constitutes a personal-data breach within the scope of the LGPD.**
The exfiltrated data (names, CPF, wallet addresses) is personal data; in a real environment this would trigger obligations including risk assessment and potential notification to the national authority (ANPD) and affected data subjects.

14. Limitations

Each limitation below is stated together with its concrete impact on the report's conclusions, so that its practical significance — rather than only its existence — is clear.

- **No hardware write blocker.** The investigation used a virtualized environment without a physical write blocker. Integrity was maintained through read-only mounts and hash verification, but physical write blocking was not employed. *Impact:* in a real proceeding this would likely draw scrutiny; here it is mitigated by the dual-hash verification (Section 6) showing the acquired image matches its stored hash, and by every subsequent analysis command shown in this report being run against read-only-mounted evidence.
- **Shared-folder acquisition instability.** The initial acquisition destination (VirtualBox shared folder) proved unstable for sustained writes; acquisition was completed to a dedicated local disk. *Impact:* none on the conclusions — the failed partial files were discarded before the successful acquisition began, and the final image's hash was independently verified twice (Section 6). This is a lab-infrastructure limitation, not an evidentiary one.
- **Process-creation auditing disabled.** Event ID 4688 was not enabled in the target's default configuration. *Impact:* removes one potential corroboration source for SDelete/wevtutil execution timing, but does not weaken any conclusion — both tools' execution is independently confirmed via Prefetch and command history (both classified as reliable in Section 10), so no finding in this report depends on the missing 4688 events.
- **Memory dump hashed with SHA-256 only.** No MD5 was generated for the memory image at acquisition time. *Impact:* minimal — SHA-256 alone is cryptographically sufficient for integrity verification; the omission is a process-consistency gap (the disk image has both hashes) rather than a reliability gap in the memory evidence itself.
- **Controlled lab environment.** Evidence was generated by the examiner in a single-investigator lab. *Impact:* the findings demonstrate methodological competence rather than analysis of independently sourced, adversarial evidence; conclusions about *this* incident are unaffected, but the exercise cannot speak to how the same methodology would perform against evidence the examiner did not generate.
- **Scope of technical evidence.** As noted in Section 11, the analysis establishes acts and attribution but not motive, intent, or downstream use of the exfiltrated data. *Impact:* directly bounds Conclusions 1–5 (Section 13) to what the evidence can support — acts, timing, and attribution — and explains why no conclusion in this report makes any claim about attacker motive or intent.

15. Recommendations

The following recommendations address both the investigative process and the defensive posture that would improve detection and prevention of comparable incidents:

1. **Enable process-creation auditing (Event ID 4688) with command-line logging.** This would have captured SDelete and wevtutil execution directly in the Security log, providing an additional independent source.
2. **Forward logs to a remote SIEM in real time.** Local log clearing (wevtutil) is only effective because logs reside solely on the host. Real-time forwarding renders local clearing ineffective and would preserve the pre-clear Security events destroyed in this incident.
3. **Restrict local administrator rights and virtual-disk creation.** Both the log clearing and the VHD creation required elevation; constraining administrative rights on analyst workstations raises the barrier to these actions.
4. **Deploy DLP and removable-media controls.** Data-loss-prevention monitoring of KYC data locations, and policy controls on mounting removable or virtual disks, would detect or prevent the exfiltration channel used here.
5. **Use hardware write blockers and dedicated evidence storage.** In any production investigation, acquire through a certified write blocker and write evidence to dedicated, verified storage rather than shared folders.
6. **Generate dual hashes (MD5 + SHA-256) for every acquired artifact at acquisition time,** including memory images.

16. Examiner Declaration

I, Paulo Vaz, hereby declare that:

1. This report accurately reflects the findings of the digital forensic examination conducted on the evidence described herein.
2. The examination was performed using recognized forensic tools and methodologies aligned with ISO/IEC 27037, 27035, 27041, 27042, 27043; NIST SP 800-86 and 800-61; and RFC 3227.
3. All procedural deviations from standard forensic practice have been documented, justified, and disclosed within this report and the chain of custody.
4. The findings, conclusions, and opinions expressed are my own and reflect an honest assessment of the evidence examined.
5. This report was prepared for academic portfolio purposes and does not constitute a legal instrument in any jurisdiction.

Paulo Vaz

Digital Forensic Examiner

Date: _____ / _____ / 2026

Generated July 12, 2026. Evidence image MD5: 7e5404e298b1d886534a2845e4d880bc. Case reference: ITI-2026-001. The NexChain Exchange organization, user accounts, and all customer data referenced are fictitious; this is an academic portfolio exercise.